# Gluconeogenesis

## The de novo synthesis of glucose and its role in preventing hypoglycemia

# Objectives

At the end of this lecture you should be able to:

- Justify the importance of gluconeogenesis in glucose homeostasis
- Describe the pathway of gluconeogenesis
- Explain how plasma glucose concentration is maintained in fed and fasting states
- Describe the relationship to glycolysis

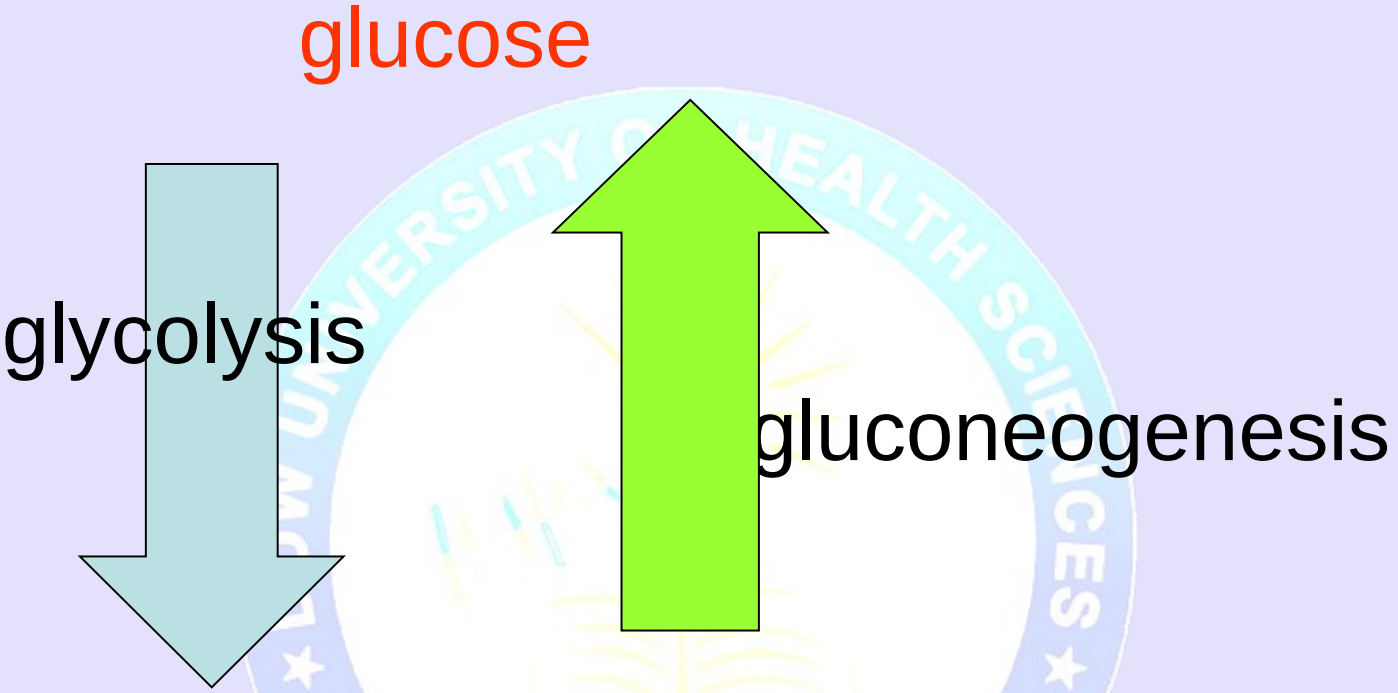
# Gluoconeogenesis

- Formation of glucose from noncarbohydrate sources e.g. lactic acid ,amino acids , glycerols and propionate.

  Site: liver and kidney.

- **Liver**
  - Approximately 90% of gluconeogenesis occurs in the liver during an overnight fast.
- **Kidneys**

Contributes up to 40% of total glucose production in fasting and more in starvation.

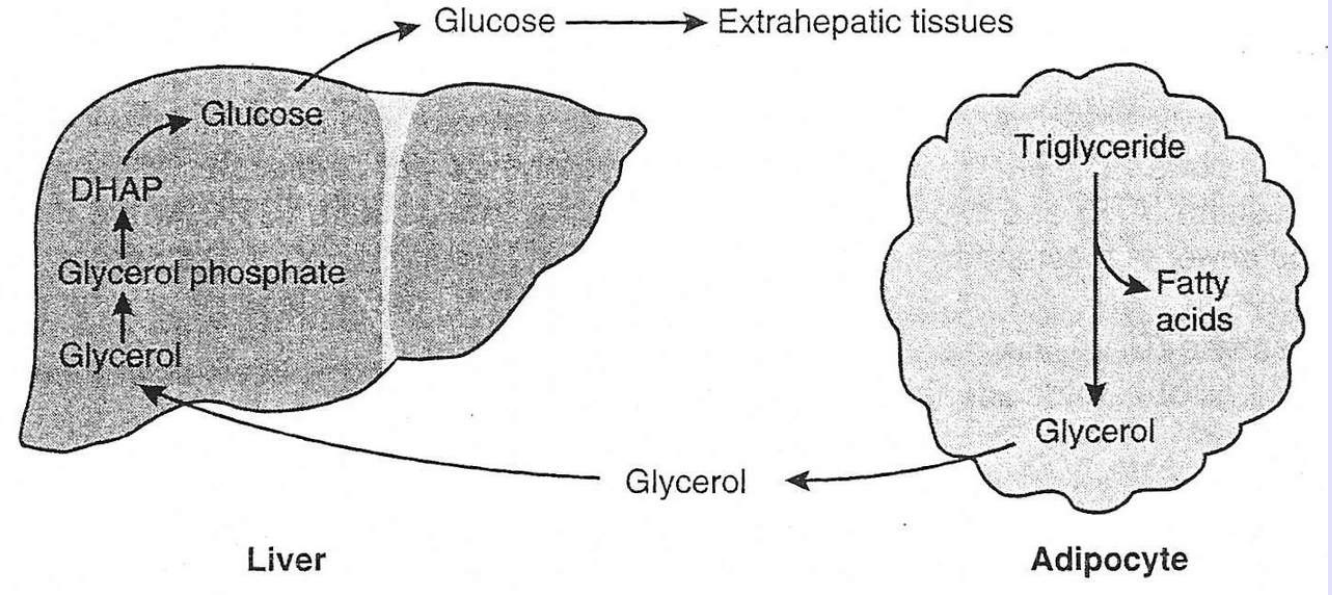



gluco neo genesis

sugar (re)new make/ create

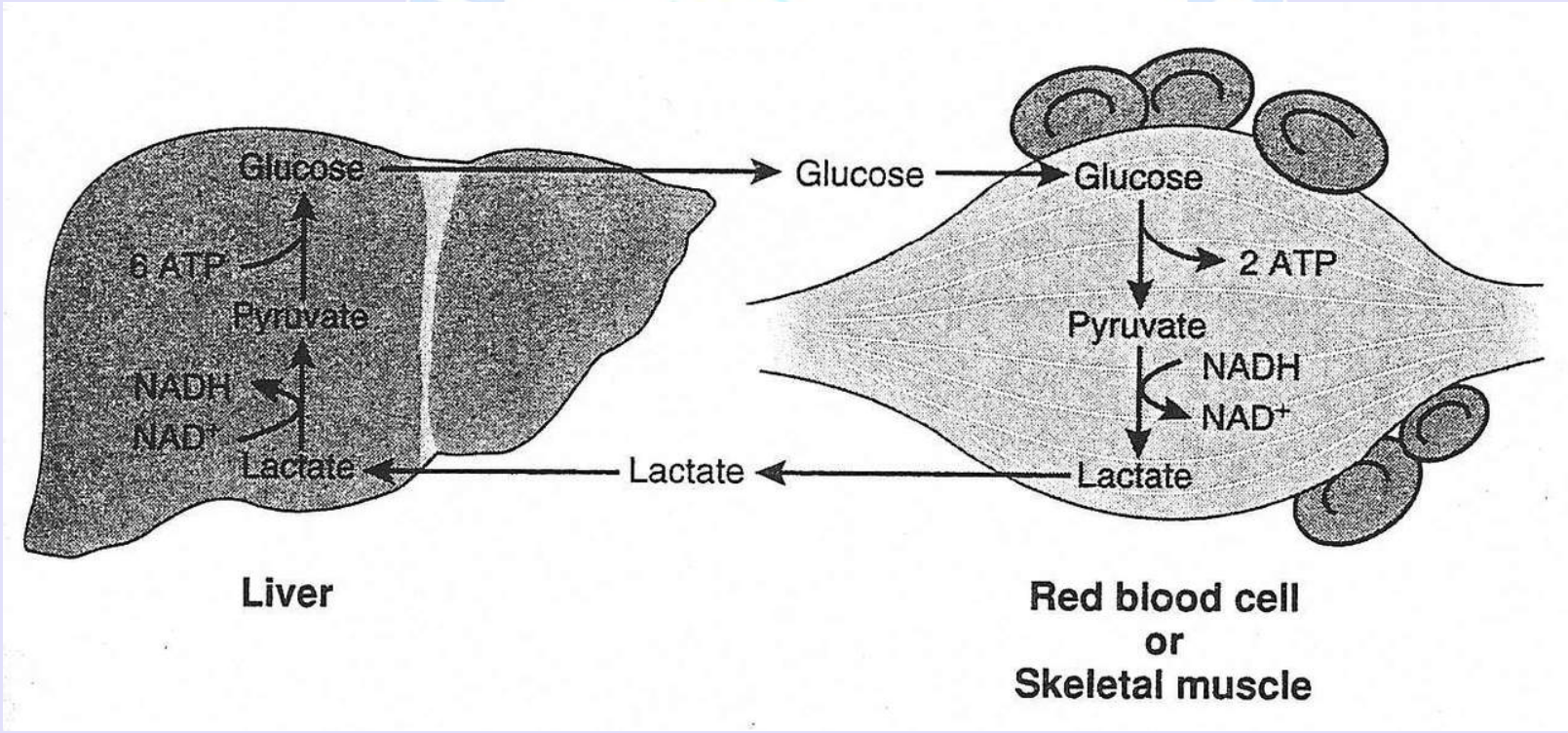
# Precursors for gluconeogenesis

- Glycerol
  - derived from adipocyte lipolysis
  - hepatic glycerol kinase



# Precursors for gluconeogenesis

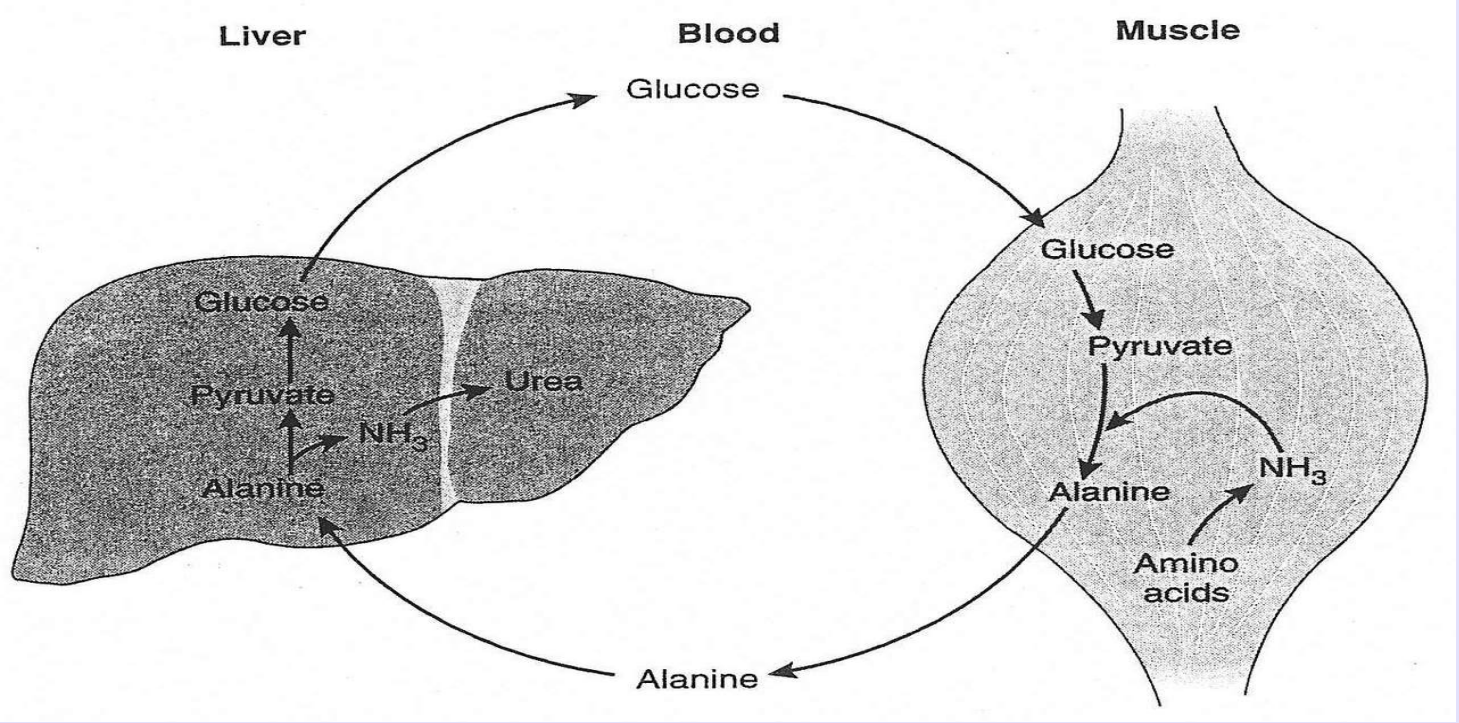
- Lactate
  - RBC
  - muscle
  - the Cori Cycle



# Precursors for gluconeogenesis

**Glucogenic amino acids:** major sources of glucose during a fast.

- derived from hydrolysis of tissue proteins e.g. Ala, Ser, Gly,Thr.
- transamination of pyruvate derived from glycolysis or from amino acid degradation.



# Importance of Gluconeogenesis

- Glucose is the only source of energy:
  - nervous system
  - Skeletal system
- Glucose is required :
  - Adipose tissues: as a source of glycerol
  - Mammary gland: as a source of lactose

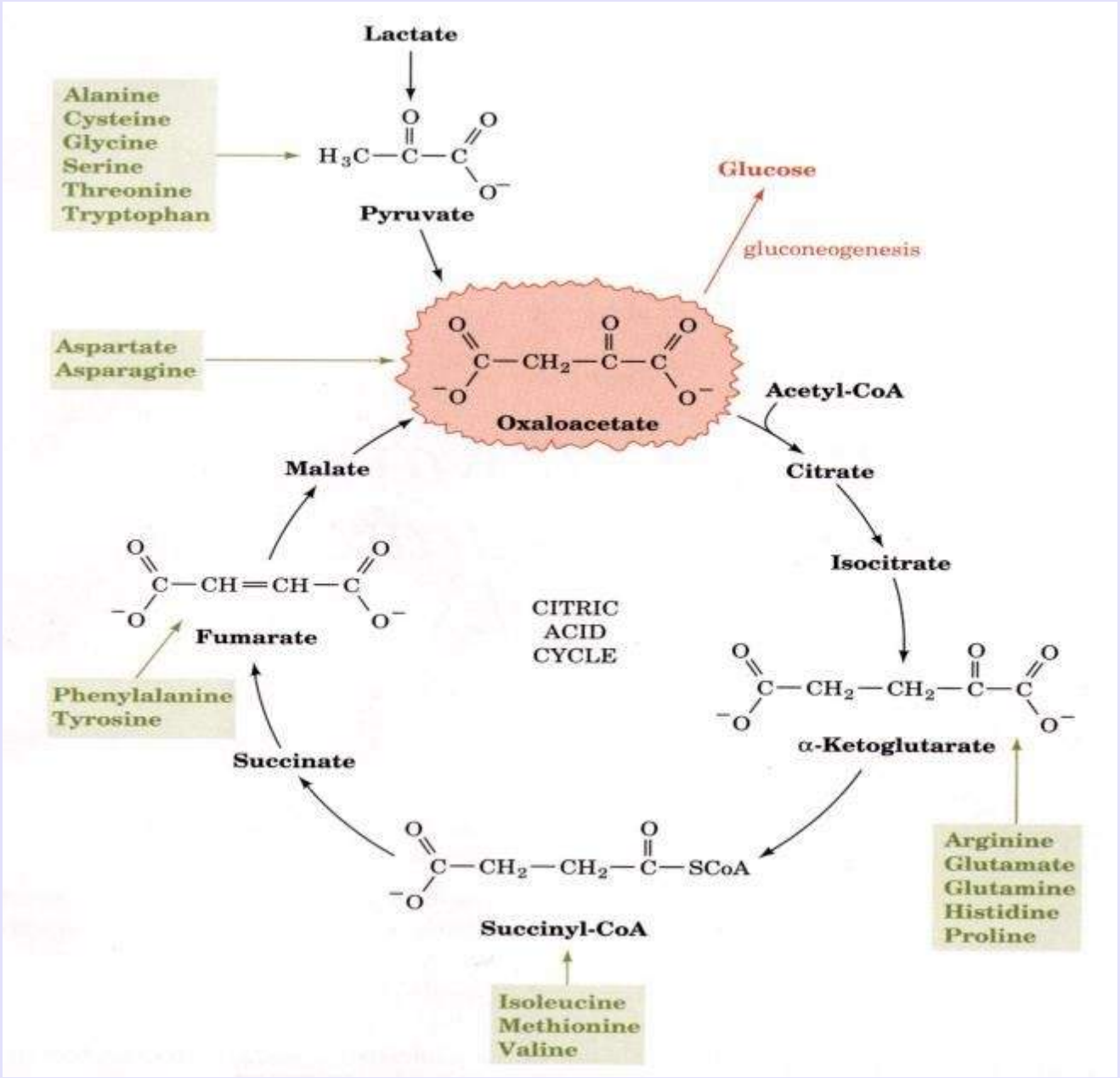
# Why is glucose important?

- Brain, red blood cells, kidney medulla, lens and cornea of the eye etc. require a continuous supply of glucose as a metabolic fuel.
- *Hypoglycemia* causes brain dysfunction, which can lead to coma and death.
- Glucose is also important in maintaining the level of intermediates of the citric acid cycle even when fatty acids are the main source of acetyl-CoA in the tissues.

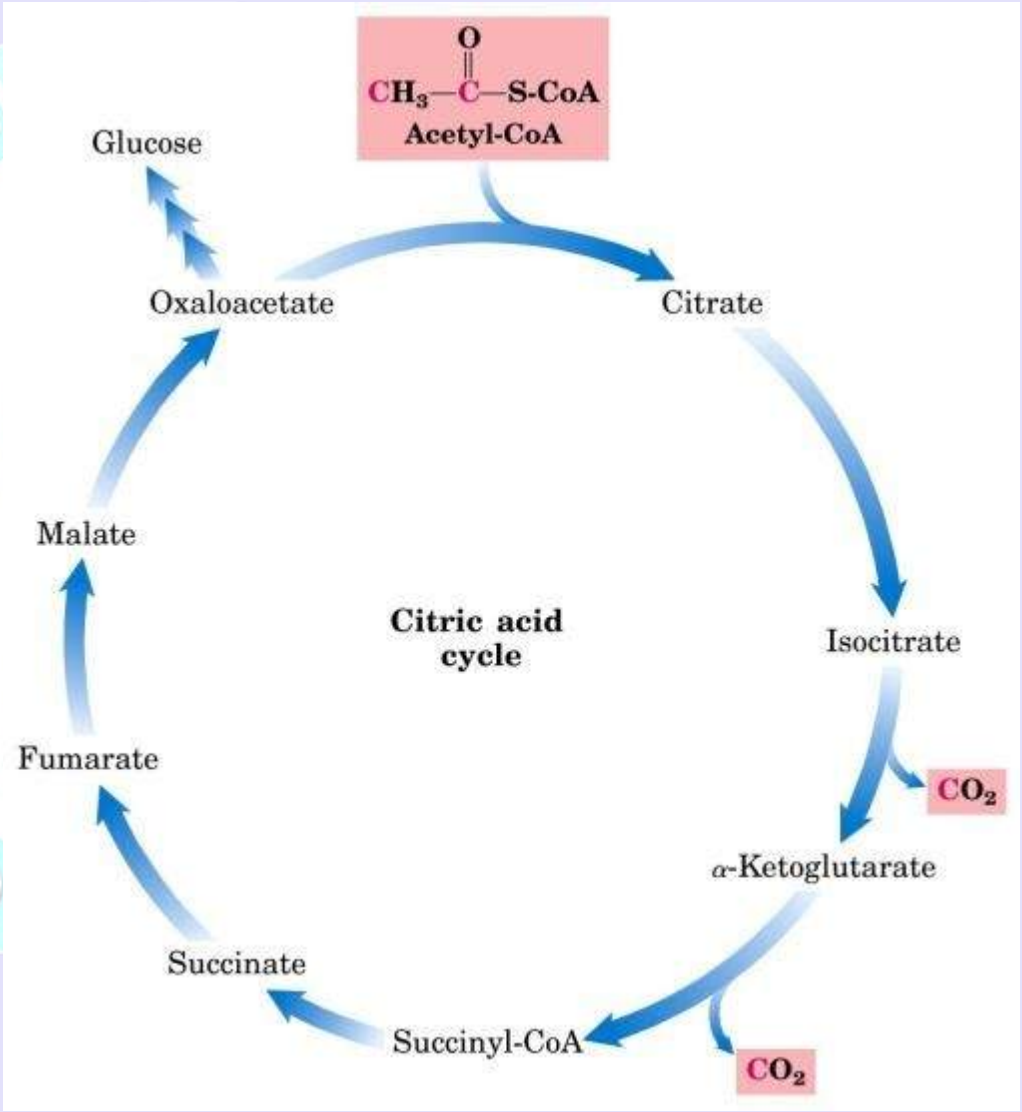
# Gluconeogenesis

The precursors such as lactate, pyruvate, glycerol, and amino acids enter the Krebs cycle at some point to be converted to oxaloacetate.

Oxaloacetate is the starting material for gluconeogenesis


Lactate
Alanine
Cysteine
Glycine
Serine
Threonine
Tryptophan
Pyruvate
Glucose
gluconeogenesis
Aspartate
Asparagine
Oxaloacetate
Acetyl-CoA
Malate
Citrate
Isocitrate
CITRIC
ACID
CYCLE
Fumarate
Phenylalanine
Tyrosine
α-Ketoglutarate
Succinate
Arginine
Glutamate
Glutamine
Histidine
Proline
Succinyl-CoA
Isoleucine
Methionine
Valine

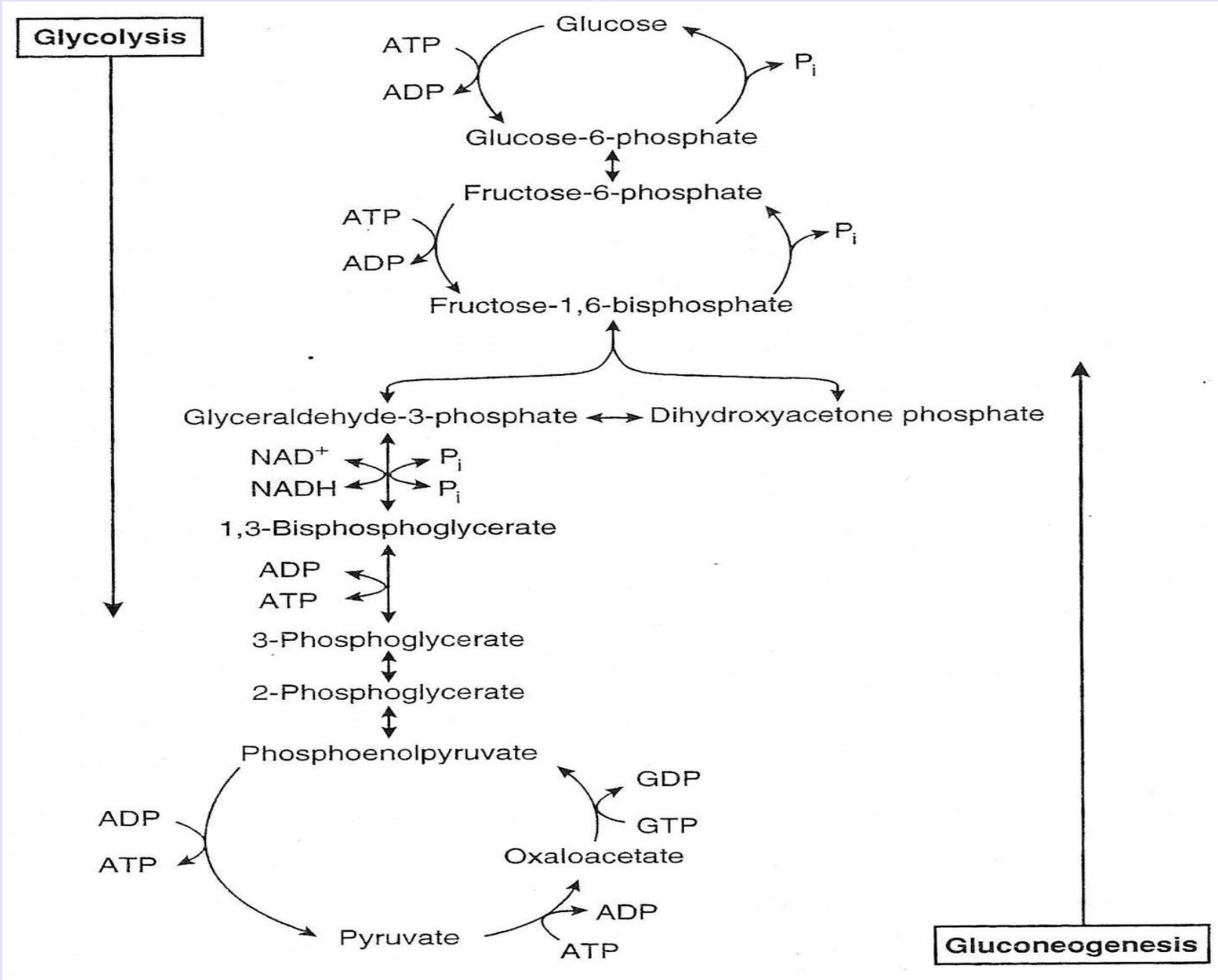
TCA intermediates
are gluconeogenic;
funnel
through oxaloacetate



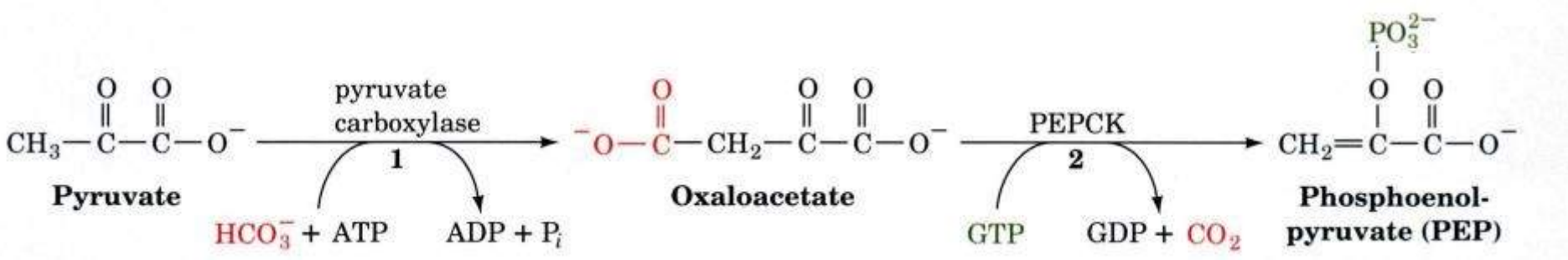
# Gluconeogenesis is not just the reverse of glycolysis

- Several steps are different so that control of one pathway does not inactivate the other. However many steps are the same. Three steps are different from glycolysis.
- 1 Pyruvate to PEP
- 2 Fructose 1,6- bisphosphate to Fructose-6-phosphate
- 3 Glucose-6-Phosphate to Glucose

# Gluconeogenesis: Overview



# Pyruvate is converted to oxaloacetate before being changed to Phosphoenolpyruvate




1. Pyruvate carboxylase catalyses the ATP-driven formation of oxaloacetate from pyruvate and $CO_2$

2. PEP carboxykinase (PEPCK) convert oxaloacetate to PEP that uses GTP as a phosphorylating agent.

**Pyruvate carboxylase requires biotin as a cofactor**

# Acetyl-CoA regulates pyruvate carboxylase

- Increase in oxaloacetate concentrations increases the activity of the Krebs cycle and acetyl-CoA is an allosteric activator of the carboxylase. However when ATP and NADH concentrations are high and the Krebs cycle is inhibited, oxaloacetate goes to form glucose.

# Synthesis of glucose from pyruvate

Synthesis of glucose from pyruvate utilizes many of the same enzymes as **Glycolysis**.

Three Glycolytic reactions are essentially **irreversible**.

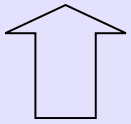
- **Hexokinase** (or Glucokinase)
- **Phosphofructokinase**
- **Pyruvate Kinase.**

These steps must be **bypassed** in Gluconeogenesis.

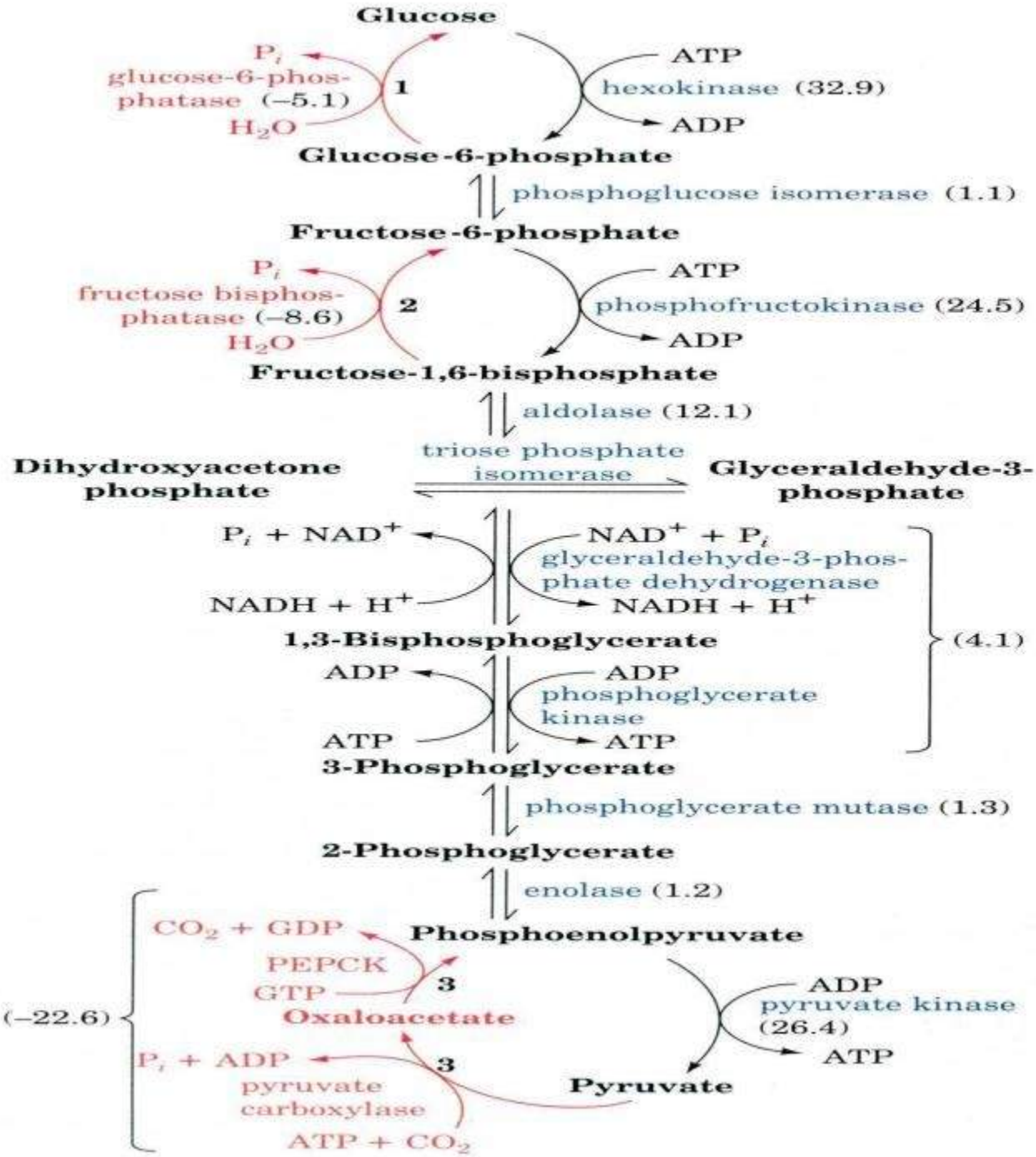
**Two** of the bypass reactions involve simple **hydrolysis** reactions.

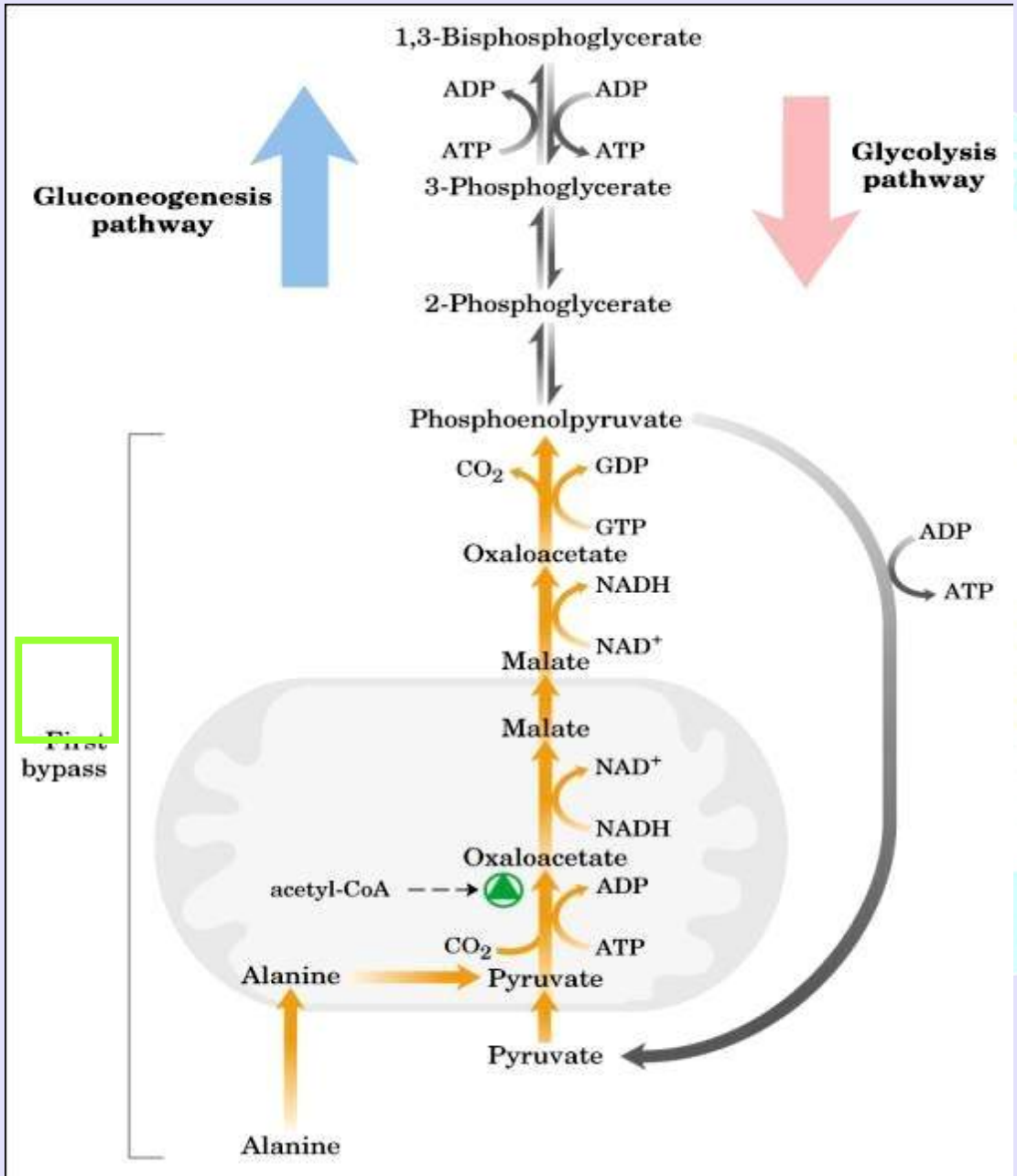
# Irreversible glycolytic steps bypassed

| glycolysis | gluconeogenesis |
| --- | --- |
| 1. Hexokinase (hexK) | by Glucose-6-phosphatase |
| 2. Phosphofructokinase-1 (PFK-1) | by Fructose 1,6-bisphosphatase (FBP-1) |
| 3. Pyruvate kinase (PyrK) | by Pyruvate Carboxylase & Phosphoenolpyruvate carboxykinase (PEPCK) |



**These 3 key enzymes**


Glucose
$P_i$
glucose-6-phos-
phatase (–5.1)
$H_2O$
1
ATP
hexokinase (32.9)
ADP
Glucose-6-phosphate
phosphoglucose isomerase (1.1)
Fructose-6-phosphate
$P_i$
fructose bisphos-
phatase (–8.6)
$H_2O$
2
ATP
phosphofructokinase (24.5)
ADP
Fructose-1,6-bisphosphate
aldolase (12.1)
Dihydroxyacetone
phosphate
triose phosphate
isomerase
Glyceraldehyde-3-
phosphate
$P_i$ + $NAD^+$
NADH + $H^+$
$NAD^+$ + $P_i$
glyceraldehyde-3-phos-
phate dehydrogenase
NADH + $H^+$
1,3-Bisphosphoglycerate
(4.1)
ADP
ATP
ADP
phosphoglycerate
kinase
ATP
3-Phosphoglycerate
phosphoglycerate mutase (1.3)
2-Phosphoglycerate
enolase (1.2)
$CO_2$ + GDP
PEPCK
GTP
3
Oxaloacetate
Phosphoenolpyruvate
ADP
pyruvate kinase
(26.4)
ATP
(–22.6)
$P_i$ + ADP
pyruvate
carboxylase
ATP + $CO_2$
3
Pyruvate




# First bypass step is generation of PEP from pyruvate via oxaloacetate

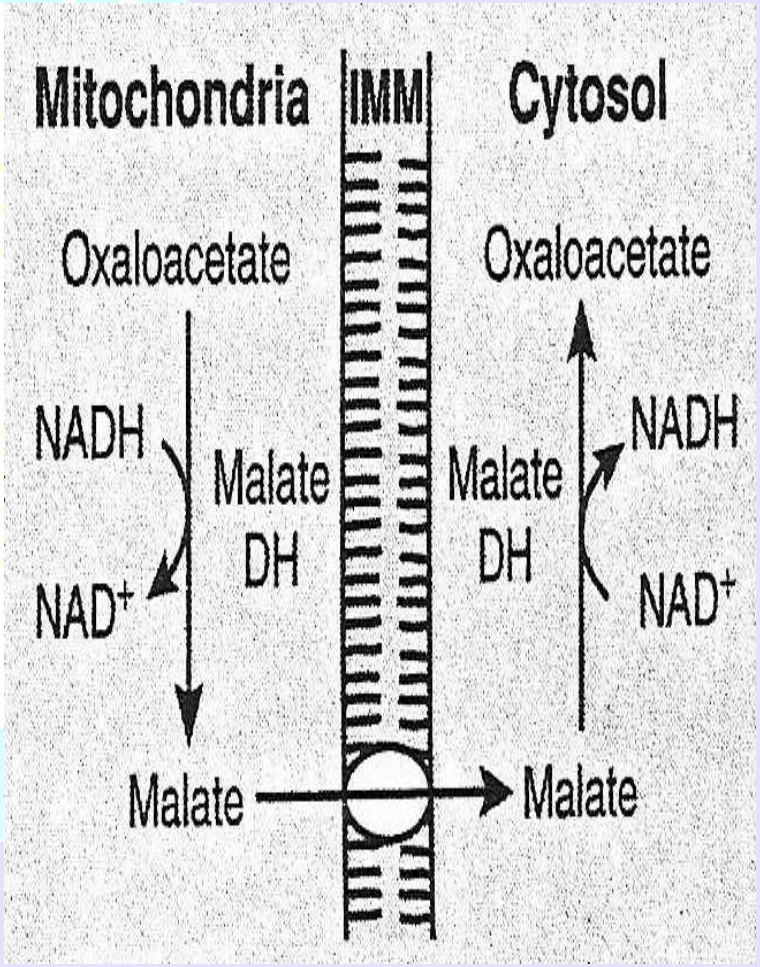
*Note:

In order to cross the mito membrane, oxaloacetate must:

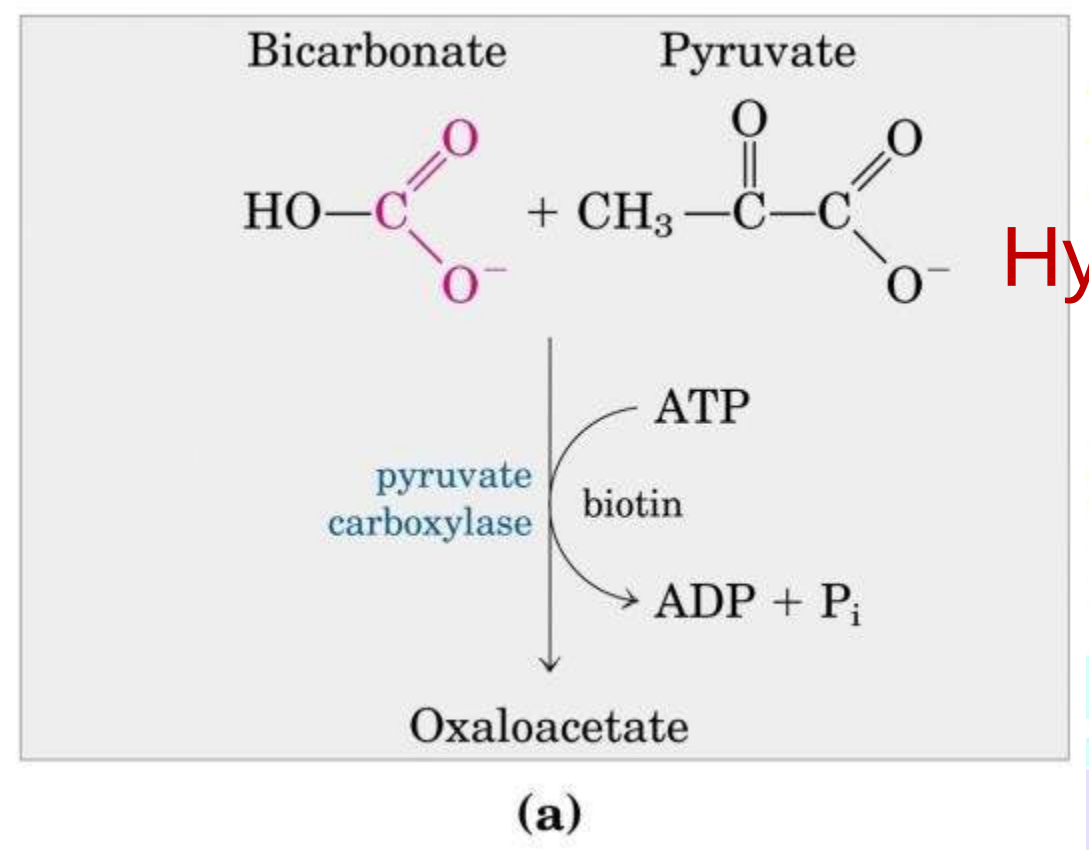
1. Be reduced to malate
2. Go through the malate shuttle
3. Be reoxidized to oxaloacetate

# Malate Shuttle

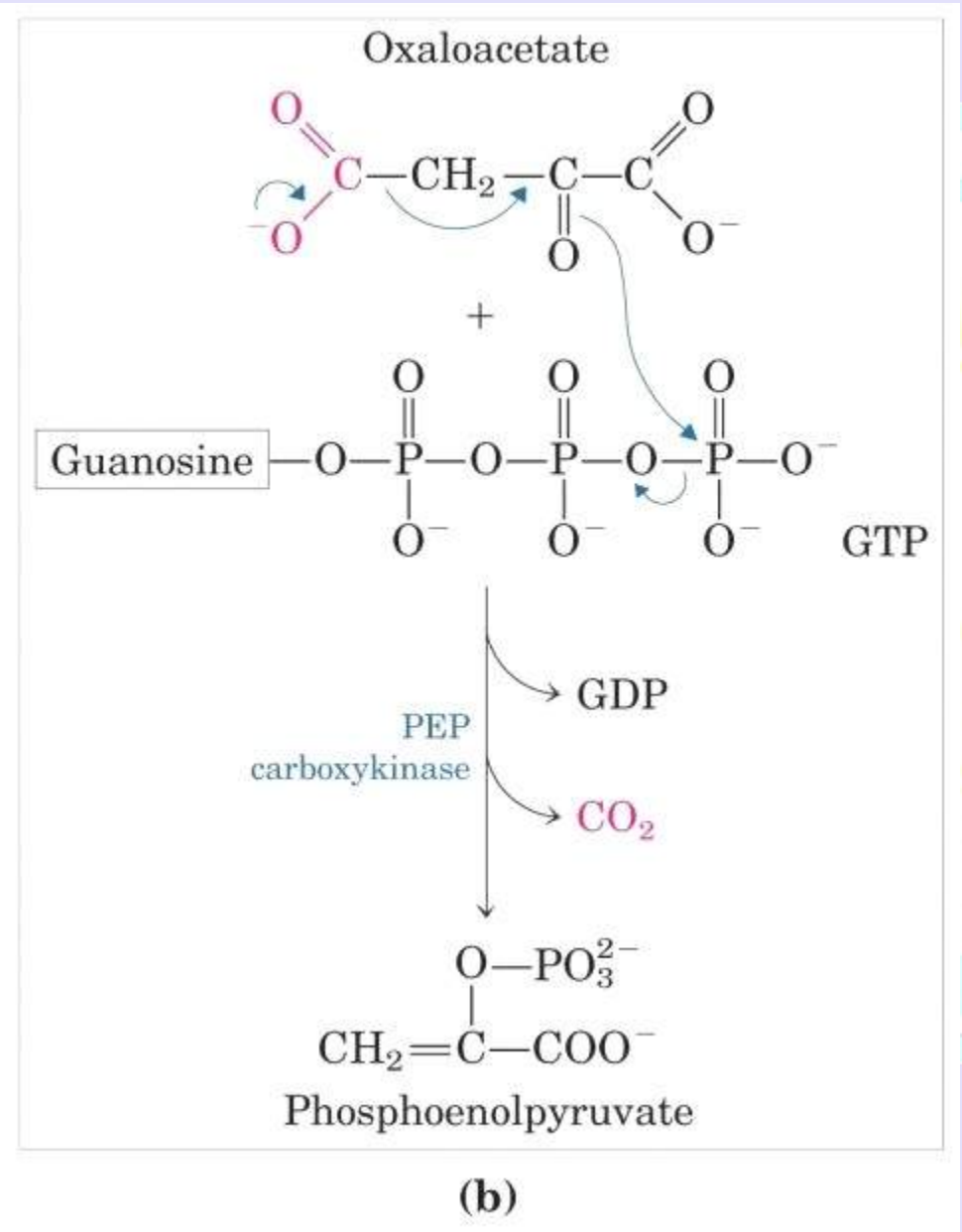
- OAA produced in mitochondria
- mitochondrial membrane impermeable to OAA
- malate transporter in mito. Membrane
- malate dehydrogenase in both mito and cyto
- NADH produced in cyto also used in gluconeogenesis.



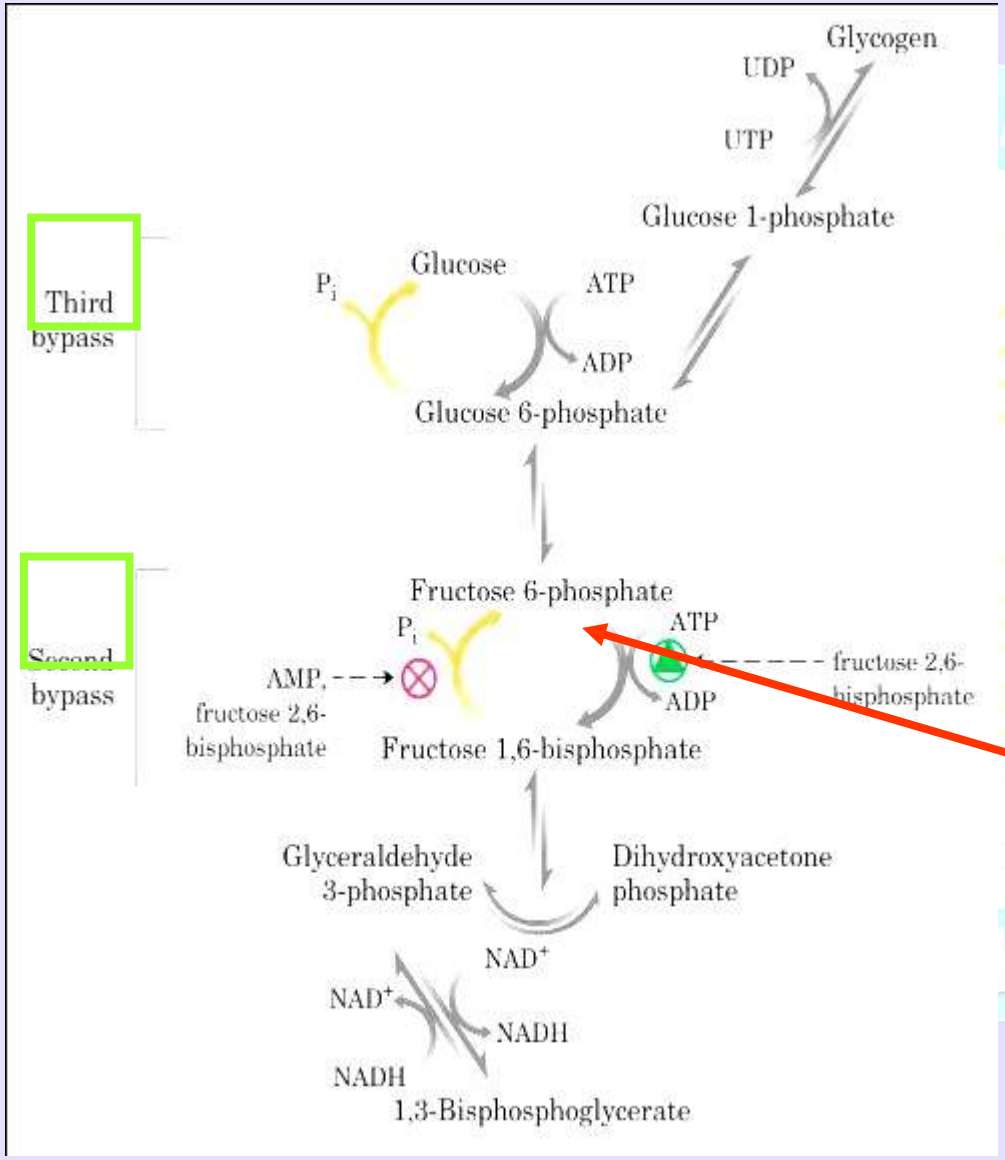
# Addition of $CO_2$ to pyruvate to form oxaloacetate




Hydrolysis of ATP




# Decarboxylation and phosphorylation to PEP




2nd & 3rd bypass steps are near the end of gluconeogenesis ("top" of glycolysis

Regulation of FBP-1 by AMP and F2,6P

# Dephosphorylation of G6P, 3rd bypass reaction

# Glucose 6-phosphatase removes the phosphate to liberate free glucose

$$\text{glucose-6-P} + H_2O \xrightarrow{\text{G6Pase}} \text{glucose} + P_i$$

- **This is primarily a function of the liver to buffer blood glucose levels**
- **G6Pase is NOT present in brain and muscle! (Gluconeogenesis does not occur in these tissues)**

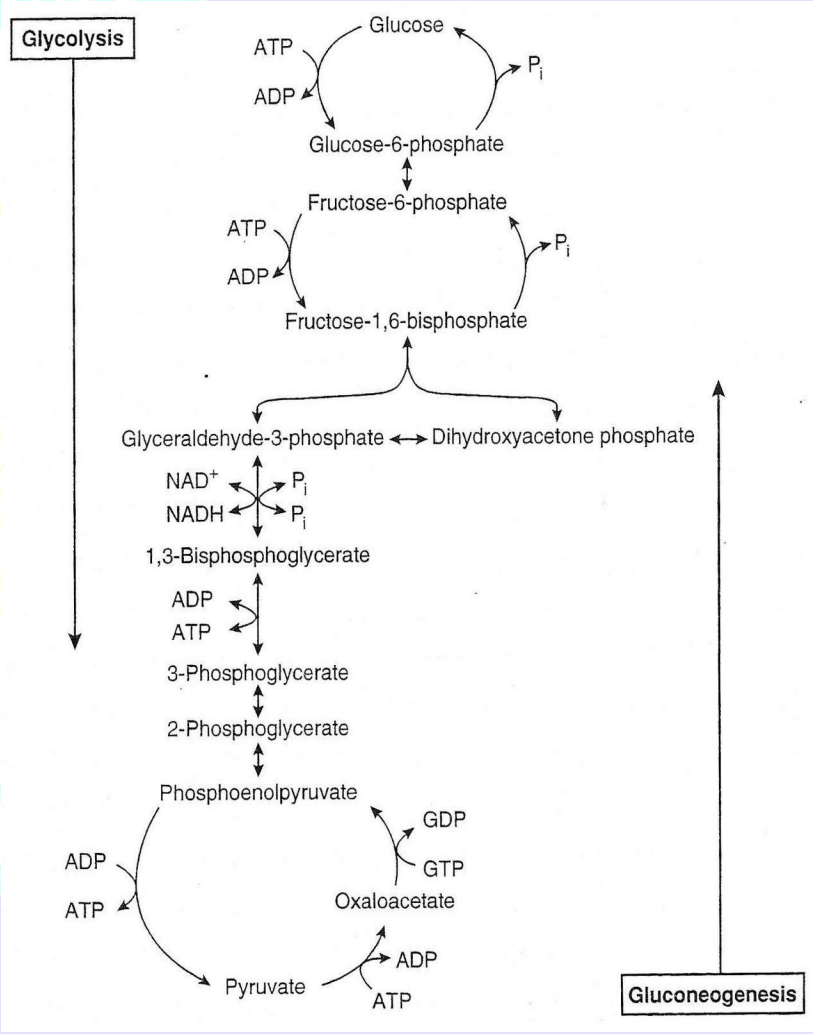
# Reactions unique to gluconeogenesis (*summary*)

Following reactions prevent reversal of glycolysis to favor the synthesis of glucose

- **Carboxylation of pyruvate**
- **Decarbooxylation and phosphorylation of oxaloacetate**
- **Dephosphorylation of fructose 1,6 bisphosphate.**
- **Dephosphorylation of Glucose 6 phosphate.**

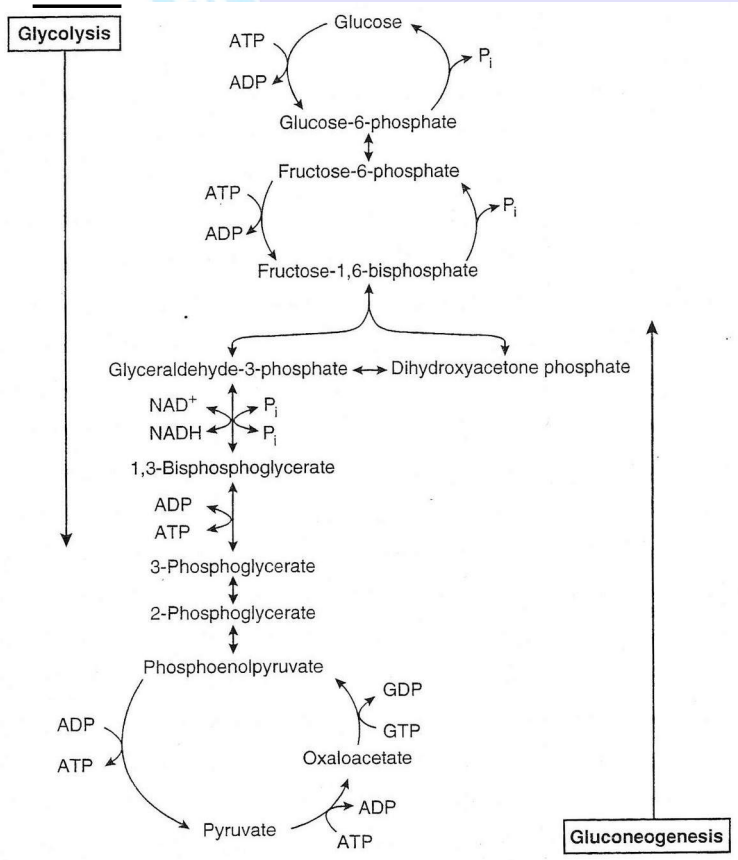
# Energetics of Gluconeogenesis

- Pyruvate Carboxylase
  - 2 ATPs
- PEP Carboxykinase
  - 2 GTPs
- 3-P-glycerate kinase
  - 2 ATPs
- Glyceraldehyde-3-P dehydrogenase
  - 2NADH



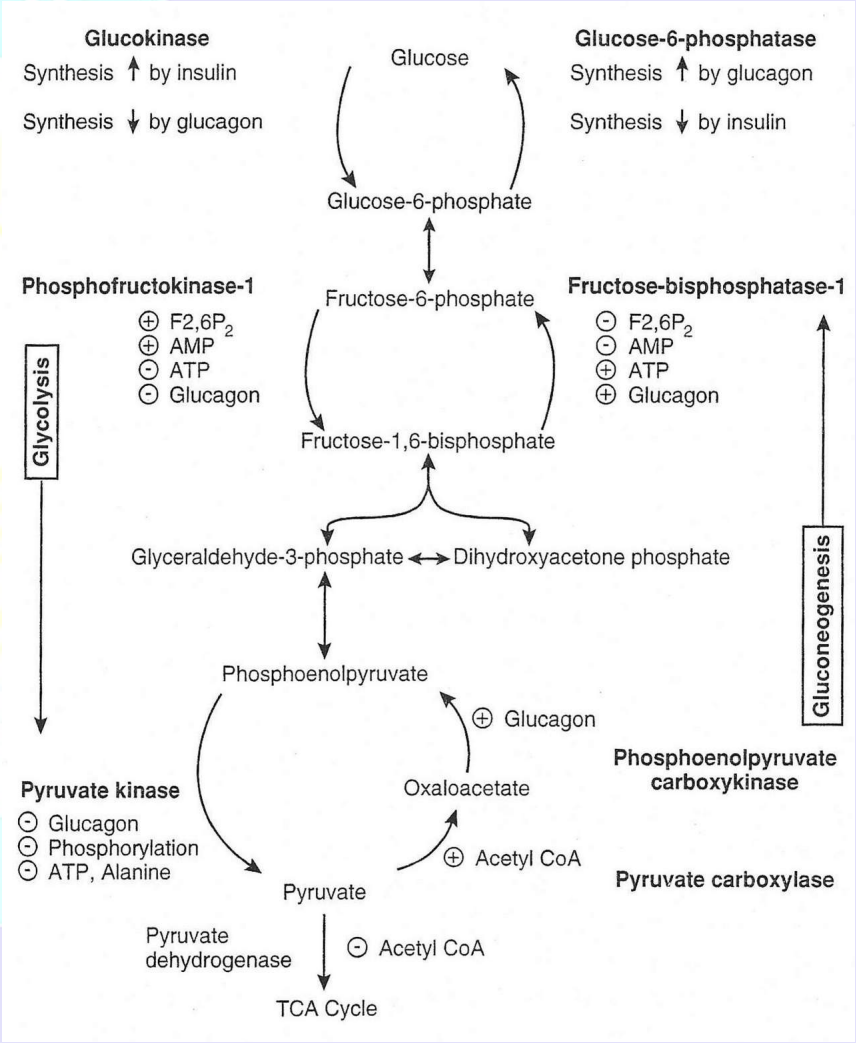
# Coordinated Regulation of Gluconeogenesis and Glycolysis

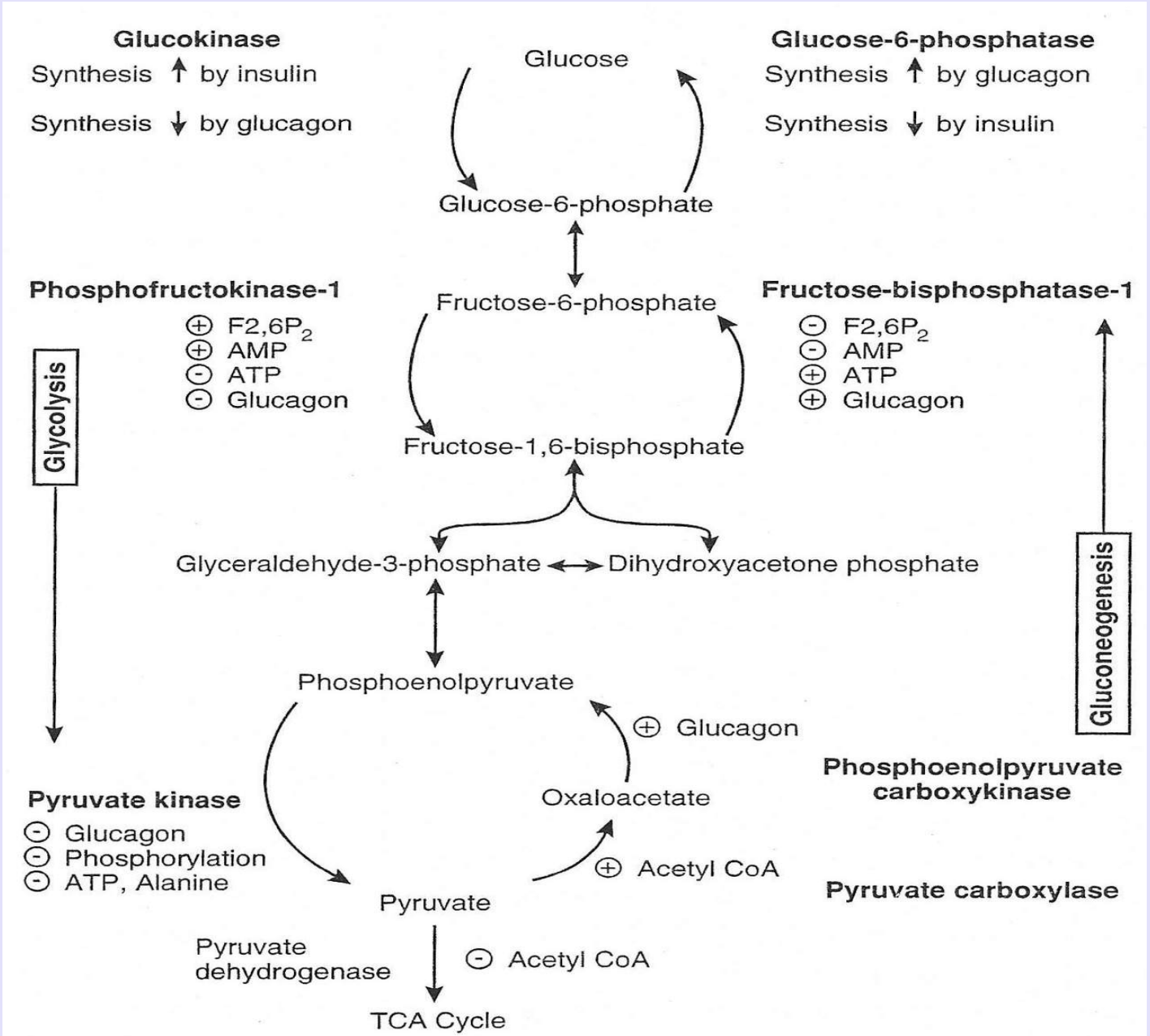
- Gluconeogenesis and Glycolysis are regulated by similar effector molecues <u>but in the opposite direction</u>
  - avoid futile cycles
    - PK vs PC&PEPCK
    - PFK-1 vs FDP'tase
    - GK vs G6P'tase



# Coordinated Regulation of Gluconeogenesis and Glycolysis

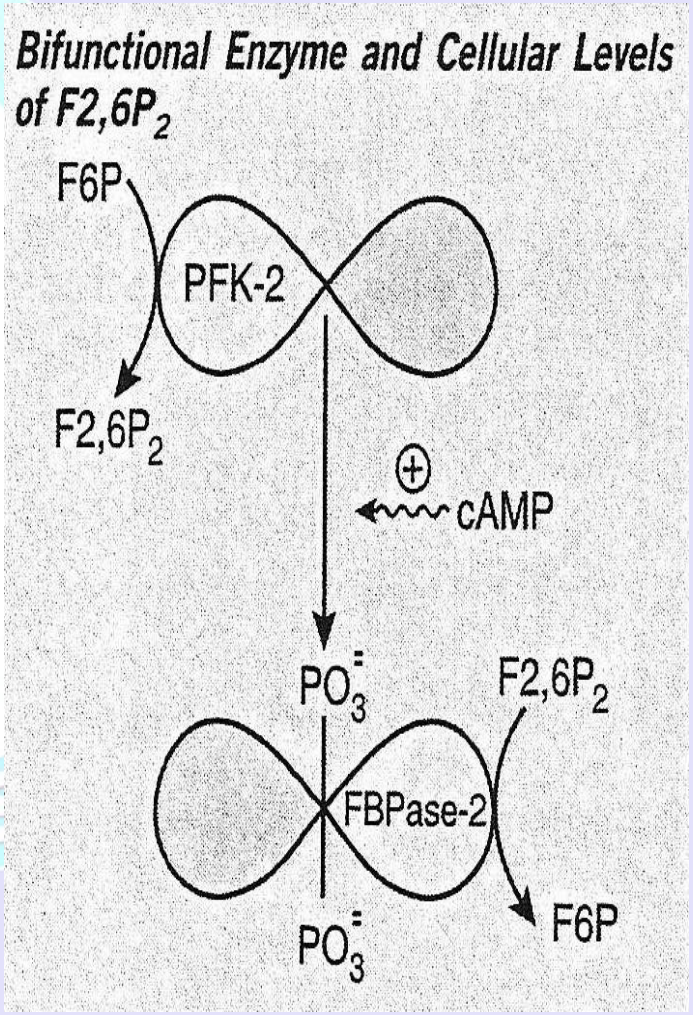
- Regulation of enzyme quantity
- Fasting: glucagon, cortisol
  - induces gluconeogenic enzymes
  - represses glycolytic enzymes
  - liver making glucose
- Feeding: insulin
  - induces glycolytic enzymes
  - represses gluconeogenic enzymes
  - liver using glucose




Glucokinase
Synthesis ↑ by insulin
Synthesis ↓ by glucagon
Glucose
Glucose-6-phosphatase
Synthesis ↑ by glucagon
Synthesis ↓ by insulin
Glucose-6-phosphate
Phosphofructokinase-1
⊕ F2,6P2
⊕ AMP
⊖ ATP
⊖ Glucagon
Fructose-6-phosphate
Fructose-bisphosphatase-1
⊖ F2,6P2
⊖ AMP
⊕ ATP
⊕ Glucagon
Glycolysis
Fructose-1,6-bisphosphate
Glyceraldehyde-3-phosphate ↔ Dihydroxyacetone phosphate
Gluconeogenesis
Phosphoenolpyruvate
⊕ Glucagon
Phosphoenolpyruvate carboxykinase
Pyruvate kinase
⊖ Glucagon
⊖ Phosphorylation
⊖ ATP, Alanine
Oxaloacetate
⊕ Acetyl CoA
Pyruvate carboxylase
Pyruvate
Pyruvate dehydrogenase
⊖ Acetyl CoA
TCA Cycle

# Coordinated Regulation of Gluconeogenesis and Glycolysis

- Short-term Hormonal Effects
  - Glucagon, Insulin
    - cAMP & $F2,6P_2$
- PFK-2 & FBPase-2
  - **A Bifunctional enzyme**
  - cAMP
    - Inactivates PFK-2
    - Activates FBPase-2
    - Decreases $F2,6P_2$
      - Reduces activation of PFK-1
      - Reduces inhibition of FBPase-1
- Low blood sugar results in
  - Hi gluconeogenesis
  - Lo glycolysis

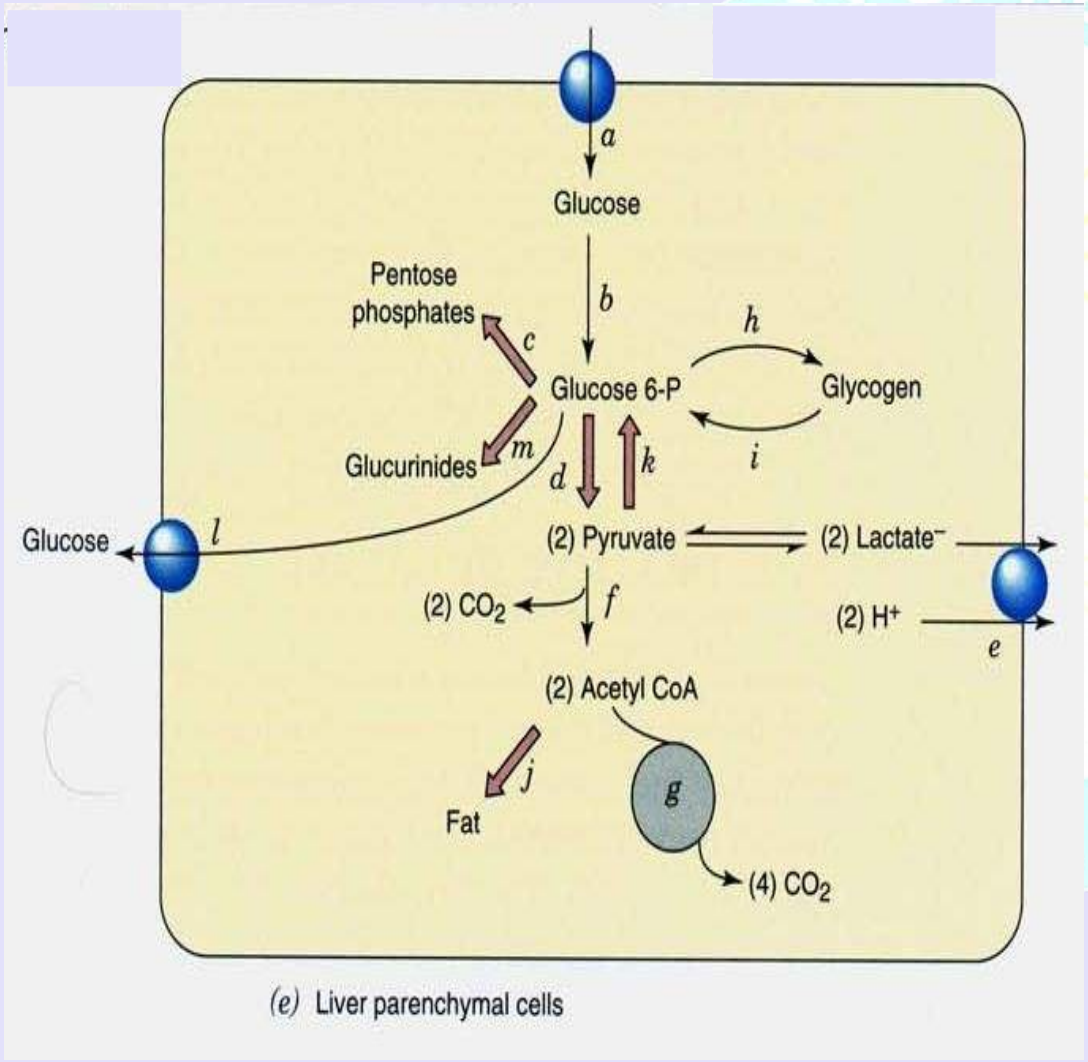


# Coordinated Regulation of Gluconeogenesis and Glycolysis

- Allosteric Effects
- Pyruvate kinase vs Pyruvate carboxylase
  - PK - Inhibited by ATP and alanine
  - PC - Activated by acetyl CoA
  - Fasting results in gluconeogenesis

- PFK-1 vs FBPase-1
  - FBPase-1 inhibited by AMP & $F2,6P_2$
  - PFK-1 activated by AMP and & $F2,6P_2$
  - Feeding results in glycolysis

# Liver is the major source of blood glucose from Gluconeogenesis (GN)



(e) Liver parenchymal cells

Is the primary gluconeogenic organ

Produces glucose for export to brain, muscle, RBC's

Uses many small metabolites and fatty acids to feed GN

Liver function is highly sensitive to insulin & glucagon

# Regulators of gluconeogenic enzyme activity

| Enzyme | Allosteric Inhibitors | Allosteric Activators | Enzyme Phosphorylation | Protein Synthesis |
|---|---|---|---|---|
| PFK | ATP, citrate | AMP, F2-6P | | |
| FBPase | AMP, F2-6P | | | |
| PK | Alanine | F1-6P | Inactivates | |
| Pyr. Carb. | | Acetyl CoA | | |
| PEPCK | | | | Glucogon |
| PFK-2 | Citrate | AMP, F6P, Pi | Inactivates | |
| FBPase-2 | F6P | Glycerol-3-P | Activates | |

DOW UNIVERSITY OF HEALTH SCIENCES D.U.H.S




The end